Pengyi Xiao

Academic Curriculum Vitae | Last updated: October 31, 2026

Intelligent Manufacturing Engineering, China University of Mining and Technology, Beijing (CUMTB)
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I am seeking MPhil or PhD opportunities starting in Fall 2028.

Research Interests

Research on applying world models to autonomous driving, embodied intelligence, and industrial production, with a focus on prediction, planning, and decision-making in dynamic environments.

Education

China University of Mining and Technology, Beijing Sep. 2024 – Jun. 2028 (expected)
B.Eng. in Intelligent Manufacturing Engineering
Tsinghua University, School of Vehicle and Mobility Aug. 2026 – Aug. 2028
Joint-Training Undergraduate Researcher (non-degree)
Formal inter-university research training agreement; Host advisor: Prof. Huang Chaosheng (Academician Li Jun's group)

Skills & Additional Training

Coursework: MITx 6.431x Probability (in progress); UC Berkeley CS285 Deep RL (self-study with assignments); Stanford EE364A Convex Optimization (selected chapters); Generative Models & Diffusion (research reading).
Technical: AutoCAD, SolidWorks, Autodesk Fusion; MATLAB, Simulink; Python, PyTorch; Linux, Git, LaTeX; C, Keil uVision 5, Altium Designer, Proteus Design Suite.
Languages: Chinese (native); English (IELTS/TOEFL score to be updated).

Research Experience

Researcher, member of Tsinghua VIN Lab, **Academician Li Jun**'s Research Group, School of Vehicle and Mobility, **Tsinghua University** Sep. 2025 – present
Role: **Researcher**
- Conduct short-horizon single-vehicle forecasting: preprocess trajectory data, extract features, validate experiments, and analyze results.
- Review related literature, document experimental protocols, and support iterative refinement of model pipelines.

Research Assistant, China Green Intelligence Urban Research Institute Jul. 2025 – Aug. 2026
Affiliated with the School of Vehicle and Mobility, **Tsinghua University**
- Organized research materials, supported data analysis, and drafted internal project documents.

Project Leader, Undergraduate Research and Innovation Project, Office of Academic Affairs, CUMTB Oct. 2025 – present
"Eye of Demeter – World-Model-Informed Energy Management for an Energy-Efficient Chicken Farm" (Advisor: Prof. Zhang Yankui)
- Designed the proposal and decomposed project tasks; coordinate the team and advisor, track milestones with weekly task lists, and prepare project presentations.
- Develop a proposed world-model approach to predict environmental and energy states and inform ventilation decisions under energy constraints.

Team Member, Undergraduate Innovation Training Program, CUMTB – Ivy-Inspired Soft Climbing Robot May 2026 – Apr. 2027 (expected)
Advisor: Prof. Li Yan Role: bio-inspired suction-cup design
- Design bio-inspired adhesion structures and multi-material attachment test plans; review soft-climbing and bio-inspired adhesion research.
- Contribute to system-integration planning for pneumatic growth, SMA actuation, retraction, and vision modules.

Research Intern, Institute of Automation, **Chinese Academy of Sciences** (Beijing) May 2025 – Jun. 2025
- Applied computer vision techniques to tooth segmentation during remote research training.

Publications and Research Outputs

Published Conference Paper

1. Zhengxian Chen, Chenggang Zhang, Yuyang Pan, Jingyue Zheng, Fojin Zhou, Peng Ni, Yichong Li, Junyi Wang, Tianyang Gong, Xun Hu, Minhao Gao, Yuxing Wu, Mingyang Li, Pengyi Xiao, Mingqiang Wang, Jun Li. **Energy Management Method for Diesel Hydrogen Hybrid Vehicles and Evaluation of Renewable Energy Applications**. *REM2025, Applied Energy Symposium and Forum*. Co-author; published.

Under Review

2. Jiwei Yang, Zhengxian Chen, Pengyi Xiao, Weihua Pan, Chenggang Zhang, Lei Yang, Wenfeng Guo, Chaosheng Huang, Jun Li. **SDRA: A Spatiotemporally Decoupled RL Agent Enhanced by a Vision-Language Model for Heavy-Duty Vehicle Eco-Driving**. *AAAI 2027*. Co-author; under review.
3. Zhengxian Chen, Han Hai, Rui Li, Pengyi Xiao, Zhengyin Gao, Chaosheng Huang, Jun Li. **The 3D Vehicle Inverted Pendulum Model: A Framework for Comfort-Oriented and Stability-Enhanced Motion Planning in Autonomous Driving**. *ICRA 2027*. Co-author; under review.

4. Jiwei Yang, Zhengxian Chen, Pengyi Xiao, Tianle Wu, Chaosheng Huang, Jun Li. **Auto-JEPA: A Latent World Model of Continuous Intent for End-to-End Autonomous Driving**. *ICLR 2027*. Co-author; under review. 🏆 Cited by Yann LeCun.
5. Zhengxian Chen, Jinhao Li, Shijia Xu, Pengyi Xiao, Lei Yang, Yanbiao Ma, Chaosheng Huang, Jun Li. **JEPA-UQ: Measuring Uncertainty in Latent World Model Rollouts**. *ICLR 2027*. Co-author; under review.

Works in Progress

- **World-Model-Based Risk Prediction and Trajectory Planning for Autonomous Driving**. Developing a framework that encodes sensor observations, ego-motion history, and navigation cues into BEV scene features; predicts future scene states; and evaluates candidate trajectories for occupancy risk, position, speed, and heading to select feasible plans.
- Zhengxian Chen, Pengyi Xiao, Boqi Li, Jiwei Yang, Lechen Hu, Jiapeng Li, Qizhang Wu, Yikai Zhang, Yanbiao Ma, Xiaoshuai Hao, Chaosheng Huang. **World Models for Autonomous Driving**. Review manuscript in preparation; co-author.
- **Patent: Hybrid-Energy Agricultural Fan Emission-Reduction System**. A CUMTB Undergraduate Research and Innovation Project combining vibration energy harvesting, environmental sensing, and world-model-informed ventilation planning.
- **Software Copyright: Agricultural Fan Environment Prediction and Regulation Assistant Software V1.0**. Application in preparation. V1.0 supports offline CSV analysis, standardized ridge-regression forecasts, and threshold-based recommendations; the world-model architecture is a proposed future extension.

Open-Source Contributions

Curator and Contributor, Awesome Autonomous Driving World Model | GitHub Repository
- Curated papers, codebases, and research resources on autonomous-driving world models for a structured, accessible literature map.

Selected Engineering and Innovation Projects

Intelligent Line-Following Car Competition Project – 1st Overall University-Wide Dec. 2024 – Apr. 2025
Role: Hardware and Debugging Lead
- Led vehicle assembly, module integration, debugging, and fault diagnosis for stable competition performance.

Energy/Water Conservation and Mathematical Modeling Competition Projects – First Prize, Beijing Apr. 2025 – present
Role: Team Leader (Overall Coordination)
- Led topic selection and task coordination; coordinated modeling and analysis, then refined the solution based on results.
- Integrated findings into the competition report and presentation and coordinated team delivery.

Biomimetic Blue Morpho Butterfly – 13th Beijing Undergraduate Mechanical Innovation Design Competition, First Prize (Beijing)
Role: Team Member

2nd CUMTB 3D Printing Creative Design Competition Finals – Third Prize Dec. 2025
Role: Team Member

Technical Team Member, *Gaitly* Team – AI Gait Correction and Scenario-Adaptive Modular Baby Walker System, "Challenge Cup" Student Entrepreneurship Competition (UIBE) 2025 – present
- Developed the technical solution for the AI gait-correction and scenario-adaptive modular system; contributed to the team's advancement from the UIBE school round.

Awards and Honours

- "Best Crisis Response" Award, Beijing International Model United Nations (BIMUN)
- First Prize, Beijing Energy/Water Conservation and Low-Carbon Emission Reduction Social Practice and Technology Competition
- First Prize (Beijing), National College Students' Mechanical Design Innovation Competition
- Second Prize (Beijing), Higher Education Cup National Undergraduate Mathematical Contest in Modeling
- Third Prize (Beijing), China International College Students' Innovation Competition
- Third Prize (Beijing), 16th "ZhengDa Cup" National College Students' Market Survey and Analysis Competition
- Third Prize (Beijing), National College Students' Financial Challenge Competition
- First Prize, University Line-Following Smart Car Competition (team ranked 1st overall)
- First Prize (College level), Career Planning Competition, School of Mechanical and Electrical Engineering (ranked 1st in college)
- 2025 Summer Social Practice: Advanced Individual; Outstanding Team; First Prize, Outstanding Practice Report
- 2026 Summer Social Practice: Advanced Individual; Outstanding Team; Third Prize, Outstanding Practice Report
- 2024–2025 CUMTB Third-Class Scholarship and Outstanding Student Cadre Recognition (certificate)

Student Leadership

Vice President, University Employment & Entrepreneurship Association Sep. 2024 – Jun. 2026
President, Model United Nations Association Sep. 2025 – Jun. 2026
Class Monitor, Intelligent Manufacturing Class 24-1 Sep. 2024 – Sep. 2025
Team Leader, Social Practice Team to Beijing, School of Mechanical and Electrical Engineering (two consecutive years) Jul. 2025 – Aug. 2026
Delegation Head, Beijing International MUN (BIMUN), Northeast China Universities MUN (NIMUN), 2025 UIBE MUN 2024 – 2025
Program Member, 2025 CUMTB (Beijing) New Youth Global Competency Talent Development Program Jun. 2025 – Mar. 2026
Event Operations Assistant, BMW Brilliance (Huachen BMW) Winter Event Feb. 2025